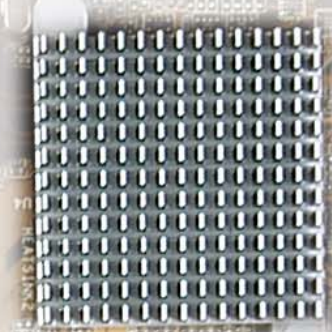
The nForce 5 series is headed by the dual-x16 lane SLI-ready 590SLI chipset with dual x16 PCI express slots—very suitable for games, or those interested in high-end processors. The nForce 570SLI is positioned as a lower-end



chipset for the gamer on a budget, and supports all the high-end features that the 590SLI does, with the exception that its PCI-Express lanes operate at half the bandwidth (x8 mode) while in SLI (multi-GPU) mode. Then there are the 570Ultra (the non-SLI 570 chipset), and the lower-end 550 and 510 chipsets.

Regular features such as the presence of additional Serial ATA ports and USB headers, expansions, etc., is never going to be a problem with motherboards based around these chipsets, simply because they're primarily high-end offerings, and so come fully loaded with almost every conceivable feature. Some manufacturers will, of course, include more features than others, which basically means a price tradeoff; others provide a stripped-down version of the same chipset, but for a much lower price. Whatever the strategy adopted, the chipset remains the same, along with all its abilities or lack of them.

nForce 5 chipsets are feature-laden just as the nForce 4s were a generation earlier, and NVIDIA's been busy adding up the goodie count on these chipsets. We received boards based on the high-end 590SLI and 570SLI chipsets.

The mid- and lower-range AM2 chipsets and motherboards based around them are scarce, and we received a solitary board based on NVIDIA's M1697, which is a mid-range solution.

Features

We received four motherboards based on nForce 590SLI chipsets—one each from ASUS, MSI, Gigabyte, and Biostar.

The ASUS Crosshair (their Premium 590SLI offering) immediately comes across as something an enthusiast would want. The board comes with an LCD display on the rear panel that gives you a POST display, helping debug any compatibility issues. Then there are the CMOS Reset, Power On/Off and Restart buttons hardwired on the board, with attractive LEDs that light them up. Great for a windowed cabinet, and a boon for enthusiasts who've spent a lot of time shorting pins and using jumpers. So heavily adorned is the PCB that ASUS couldn't provide onboard sound with the board, but it provides an 8-channel soundcard that fits into the PCI-E (x1) slot. The only gripe with all ASUS boards is their graphics card retention mechanism, which looks tacky and doesn't hold the card firmly.

The Crosshair is built around an eight-phase power design—well-endowed to stand the rigours of overclocking, and ASUS has used solid-state capacitors around the CPU area—which they call "EL-Capless" design. And overclock you can, with the option to increase the CPU multiplier ratio to 25x. You'll need an FX series processor to be able to tinker around here, as the other Athlon processors come multiplier-locked. Even the communication speed between the Northbridge and Southbridge can be tweaked all the way up to 400 MHz (200 MHz is default). The CPU bus speed can also be set to 400 MHz (default is 200 MHz). Hypothetically, a multiplier ratio of 15x with a default speed of 200 MHz will have that processor churning at 3000 MHz. Storing and loading multiple BIOS settings is also possible, and this saves you the time spent tweaking the BIOS every time you have a freeze-up. Choosing profiles is hassle free, and you can overclock whenever you startup, by simply selecting an overclocked profile.

The Biostar Tforce 590SLI has the most innovative-looking bundle we've seen in a while. All the connectors came in a black nylon casing, while the mic and headphone (single earbud type, Bluetooth-style) came in a small black box. Also present were a number of adapters with which you can charge your cell phone using any one of this board's USB ports. While not a new concept, it's commendable, and we rather liked the little extra effort Biostar put in here.

Although the board itself looks good, with flashy colours, we found that the CPU's heatsink kept interfering with the memory installed in the first DIMM. We actually had to remove the heatsink, install the memory, and then install the heatsink again! On a more technical note, we were pleased to see high-quality solid-state capacitors used liberally around the CPU socket area, instead of the electrolyte capacitors normally used on most boards.

There are dynamic overclocking profiles (V6, V8 and V12) that allow up to 30 per cent dynamic overclocking. Manual overclocking is possible as well. The Tforce had one peculiarity—memory voltage cannot be increased past 2.2 volts, unless you use a jumper on the board; and then your memory voltage will be stuck at 2.3 volts. A very peculiar trait for what is positioned as an enthusiast/overclocker board.

Gigabyte's GA-M59SLI-S5 was a pretty solid looking board, well built and reasonably laid out (not as good as the ASUS Crosshair, though). No complaints, except that larger coolers (compared to the stock AMD cooler) may interfere with the installation of memory in the first DIMM. We prefer all the connectors (24-pin, ATA and floppy) to be bunched together at one end of the board, mainly to reduce clutter from around the middle of the PCB—Gigabyte doesn't disappoint here!

This Gigabyte board also sports an attractive-looking heatpipe solution. You'll find